

1 Calcul des termes en L pour Solide et Fluide

1.1 $L = \mathcal{F}^{-1}(s_0 s_1 s_2)$

$$\begin{aligned}
 L_{abc} = & -\alpha_0 \delta_0 - \alpha_1 \delta_1 - \alpha_2 \delta_2 + \delta_0 \delta_1 + \delta_0 \delta_2 + \delta_1 \delta_2 + \omega^2 + \omega (\delta_0 + \delta_1 + \delta_2) \\
 & + \frac{\alpha_0^2 \delta_0}{(\alpha_0 - \alpha_1)(\alpha_0 - \alpha_2)} (-\alpha_0 + \alpha_1 + \delta_1) (-\alpha_0 + \alpha_2 + \delta_2) \cdot \frac{1}{\alpha_0 + \omega} \\
 & + -\frac{\alpha_1^2 \delta_1}{(\alpha_0 - \alpha_1)(\alpha_1 - \alpha_2)} (\alpha_0 - \alpha_1 + \delta_0) (-\alpha_1 + \alpha_2 + \delta_2) \cdot \frac{1}{\alpha_1 + \omega} \\
 & + \frac{\alpha_2^2 \delta_2 (\alpha_0 - \alpha_2 + \delta_0) (\alpha_1 - \alpha_2 + \delta_1)}{(\alpha_0 - \alpha_2)(\alpha_1 - \alpha_2)} \cdot \frac{1}{\alpha_2 + \omega}
 \end{aligned}$$

$$\begin{aligned}
 L_{aac} = & -\alpha_0 \delta_0 - \alpha_0 \delta_1 - \alpha_2 \delta_2 + \delta_0 \delta_1 + \delta_0 \delta_2 + \delta_1 \delta_2 + \omega^2 + \omega (\delta_0 + \delta_1 + \delta_2) \\
 & + \frac{\alpha_0}{(\alpha_0 - \alpha_2)^2} (\alpha_0^3 (\delta_0 + \delta_1) + \alpha_0^2 (-2\alpha_2 \delta_0 - 2\alpha_2 \delta_1 - 2\delta_0 \delta_1 - \delta_0 \delta_2 - \delta_1 \delta_2) + \alpha_0 (\alpha_2^2 \delta_0 + \alpha_2^2 \delta_1 + 4\alpha_2 \delta_0 \delta_1 + \alpha_2 \delta_0 \delta_2 + \alpha_2 \delta_1 \delta_2 + \delta_0 \delta_1 \delta_2) - 2\alpha_2^2 \delta_0 \delta_1 - 2\alpha_2 \delta_0 \delta_1 \delta_2) \cdot \frac{1}{\alpha_0 + \omega} \\
 & + \frac{\alpha_0^2 \delta_0 \delta_1}{\alpha_0 - \alpha_2} (\alpha_0 - \alpha_2 - \delta_2) \cdot \frac{1}{(\alpha_0 + \omega)^2} \\
 & + \frac{\alpha_2^2 \delta_2}{(\alpha_0 - \alpha_2)^2} (\alpha_0 - \alpha_2 + \delta_0) (\alpha_0 - \alpha_2 + \delta_1) \cdot \frac{1}{\alpha_2 + \omega}
 \end{aligned}$$

$$\begin{aligned}
 L_{aba} = & -\alpha_0 \delta_0 - \alpha_0 \delta_2 - \alpha_1 \delta_1 + \delta_0 \delta_1 + \delta_0 \delta_2 + \delta_1 \delta_2 + \omega^2 + \omega (\delta_0 + \delta_1 + \delta_2) \\
 & + \frac{\alpha_0}{(\alpha_0 - \alpha_1)^2} (\alpha_0^3 (\delta_0 + \delta_2) + \alpha_0^2 (-2\alpha_1 \delta_0 - 2\alpha_1 \delta_2 - \delta_0 \delta_1 - 2\delta_0 \delta_2 - \delta_1 \delta_2) + \alpha_0 (\alpha_1^2 \delta_0 + \alpha_1^2 \delta_2 + \alpha_1 \delta_0 \delta_1 + 4\alpha_1 \delta_0 \delta_2 + \alpha_1 \delta_1 \delta_2 + \delta_0 \delta_1 \delta_2) - 2\alpha_1^2 \delta_0 \delta_2 - 2\alpha_1 \delta_0 \delta_1 \delta_2) \cdot \frac{1}{\alpha_0 + \omega} \\
 & + \frac{\alpha_0^2 \delta_0 \delta_2}{\alpha_0 - \alpha_1} (\alpha_0 - \alpha_1 - \delta_1) \cdot \frac{1}{(\alpha_0 + \omega)^2} \\
 & + \frac{\alpha_1^2 \delta_1}{(\alpha_0 - \alpha_1)^2} (\alpha_0 - \alpha_1 + \delta_0) (\alpha_0 - \alpha_1 + \delta_2) \cdot \frac{1}{\alpha_1 + \omega}
 \end{aligned}$$

$$\begin{aligned}
 L_{abb} = & -\alpha_0 \delta_0 - \alpha_1 \delta_1 - \alpha_1 \delta_2 + \delta_0 \delta_1 + \delta_0 \delta_2 + \delta_1 \delta_2 + \omega^2 + \omega (\delta_0 + \delta_1 + \delta_2) \\
 & + \frac{\alpha_0^2 \delta_0}{(\alpha_0 - \alpha_1)^2} (-\alpha_0 + \alpha_1 + \delta_1) (-\alpha_0 + \alpha_1 + \delta_2) \cdot \frac{1}{\alpha_0 + \omega} \\
 & + \frac{\alpha_1}{(\alpha_0 - \alpha_1)^2} (\alpha_0^2 (\alpha_1 \delta_1 + \alpha_1 \delta_2 - 2\delta_1 \delta_2) + \alpha_0 (-2\alpha_1^2 \delta_1 - 2\alpha_1^2 \delta_2 + \alpha_1 \delta_0 \delta_1 + \alpha_1 \delta_0 \delta_2 + 4\alpha_1 \delta_1 \delta_2 - 2\delta_0 \delta_1 \delta_2) + \alpha_1^3 (\delta_1 + \delta_2) + \alpha_1^2 (-\delta_0 \delta_1 - \delta_0 \delta_2 - 2\delta_1 \delta_2) + \alpha_1 \delta_0 \delta_1 \delta_2) \cdot \frac{1}{\alpha_1 + \omega} \\
 & + \frac{\alpha_1^2 \delta_1 \delta_2}{\alpha_0 - \alpha_1} (\alpha_0 - \alpha_1 + \delta_0) \cdot \frac{1}{(\alpha_1 + \omega)^2}
 \end{aligned}$$

$$\begin{aligned}
L_{aaa} = & -\alpha_0\delta_0 - \alpha_0\delta_1 - \alpha_0\delta_2 + \delta_0\delta_1 + \delta_0\delta_2 + \delta_1\delta_2 + \omega^2 + \omega(\delta_0 + \delta_1 + \delta_2) \\
& + \alpha_0^2(\delta_0 + \delta_1 + \delta_2) + \alpha_0(-2\delta_0\delta_1 - 2\delta_0\delta_2 - 2\delta_1\delta_2) + \delta_0\delta_1\delta_2 \cdot \frac{1}{\alpha_0 + \omega} \\
& + \alpha_0^2\delta_0\delta_1\delta_2 \cdot \frac{1}{(\alpha_0 + \omega)^3} \\
& + \alpha_0(\alpha_0(\delta_0\delta_1 + \delta_0\delta_2 + \delta_1\delta_2) - 2\delta_0\delta_1\delta_2) \cdot \frac{1}{(\alpha_0 + \omega)^2}
\end{aligned}$$

1.2 $L = \mathcal{F}^{-1}(s_0 s_1)$

$$\begin{aligned}
L_{ab} = & -\alpha_0\delta_0 - \alpha_1\delta_1 - \alpha_2\delta_2 + \delta_0\delta_1 + \delta_0\delta_2 + \delta_1\delta_2 + \omega^2 + \omega(\delta_0 + \delta_1 + \delta_2) \\
& + \frac{\alpha_0^2\delta_0}{(\alpha_0 - \alpha_1)(\alpha_0 - \alpha_2)}(-\alpha_0 + \alpha_1 + \delta_1)(-\alpha_0 + \alpha_2 + \delta_2) \cdot \frac{1}{\alpha_0 + \omega} \\
& + -\frac{\alpha_1^2\delta_1}{(\alpha_0 - \alpha_1)(\alpha_1 - \alpha_2)}(\alpha_0 - \alpha_1 + \delta_0)(-\alpha_1 + \alpha_2 + \delta_2) \cdot \frac{1}{\alpha_1 + \omega} \\
& + \frac{\alpha_2^2\delta_2(\alpha_0 - \alpha_2 + \delta_0)(\alpha_1 - \alpha_2 + \delta_1)}{(\alpha_0 - \alpha_2)(\alpha_1 - \alpha_2)} \cdot \frac{1}{\alpha_2 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{aa} = & -\alpha_0\delta_0 - \alpha_0\delta_1 + \delta_0\delta_1 + \omega^2 + \omega(\delta_0 + \delta_1) \\
& + \alpha_0(\alpha_0(\delta_0 + \delta_1) - 2\delta_0\delta_1) \cdot \frac{1}{\alpha_0 + \omega} \\
& + \alpha_0^2\delta_0\delta_1 \cdot \frac{1}{(\alpha_0 + \omega)^2}
\end{aligned}$$

2 Calcul des termes L_{ijk}

2.1 $L^{012} = \mathcal{F}^{-1}\left(\frac{s_0 s_1}{s_2}\right)$

$$\begin{aligned}
L_{abc}^{012} = & 1 \\
& + \frac{\delta_0(\alpha_0 - \alpha_2)(-\alpha_0 + \alpha_1 + \delta_1)}{(\alpha_0 - \alpha_1)(-\alpha_0 + \alpha_2 + \delta_2)} \cdot \frac{1}{\alpha_0 + \omega} \\
& + -\frac{\delta_1(\alpha_1 - \alpha_2)(\alpha_0 - \alpha_1 + \delta_0)}{(\alpha_0 - \alpha_1)(-\alpha_1 + \alpha_2 + \delta_2)} \cdot \frac{1}{\alpha_1 + \omega} \\
& + -\frac{\delta_2(\alpha_0 - \alpha_2 + \delta_0 - \delta_2)(\alpha_1 - \alpha_2 + \delta_1 - \delta_2)}{(-\alpha_0 + \alpha_2 + \delta_2)(-\alpha_1 + \alpha_2 + \delta_2)} \cdot \frac{1}{\alpha_2 + \delta_2 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{aac}^{012} = & 1 \\
& + \frac{1}{(-\alpha_0 + \alpha_2 + \delta_2)^2} (\alpha_0^2 (\delta_0 + \delta_1) + \alpha_0 (-2\alpha_2\delta_0 - 2\alpha_2\delta_1 - \delta_0\delta_2 - \delta_1\delta_2) + \alpha_2^2 (\delta_0 + \delta_1) + \alpha_2 (\delta_0\delta_2 + \delta_1\delta_2) + \delta_0\delta_1\delta_2) \cdot \frac{1}{\alpha_0 + \omega} \\
& + \frac{\delta_0\delta_1 (\alpha_0 - \alpha_2)}{\alpha_0 - \alpha_2 - \delta_2} \cdot \frac{1}{(\alpha_0 + \omega)^2} \\
& + -\frac{\delta_2}{(-\alpha_0 + \alpha_2 + \delta_2)^2} (\alpha_0 - \alpha_2 + \delta_0 - \delta_2) (\alpha_0 - \alpha_2 + \delta_1 - \delta_2) \cdot \frac{1}{\alpha_2 + \delta_2 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{aba}^{012} = & 1 \\
& + \frac{1}{(\alpha_0 - \alpha_1)^2} (-\alpha_0^3 + \alpha_0^2 (2\alpha_1 + \alpha_2 + \delta_0 + \delta_1) + \alpha_0 (-\alpha_1^2 - 2\alpha_1\alpha_2 - 2\alpha_1\delta_0 - \alpha_1\delta_1 - \alpha_2\delta_1) + \alpha_1^2 (\alpha_2 + \delta_0) + \alpha_1 (\alpha_2\delta_1 + \delta_0\delta_1) - \alpha_2\delta_0\delta_1) \cdot \frac{1}{\alpha_0 + \omega} \\
& + \frac{\delta_0}{\alpha_0 - \alpha_1} (\alpha_0 - \alpha_2) (-\alpha_0 + \alpha_1 + \delta_1) \cdot \frac{1}{(\alpha_0 + \omega)^2} \\
& + -\frac{\delta_1}{(\alpha_0 - \alpha_1)^2} (\alpha_1 - \alpha_2) (\alpha_0 - \alpha_1 + \delta_0) \cdot \frac{1}{\alpha_1 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{abb}^{012} = & 1 \\
& + -\frac{\delta_0}{(\alpha_0 - \alpha_1)^2} (\alpha_0 - \alpha_2) (-\alpha_0 + \alpha_1 + \delta_1) \cdot \frac{1}{\alpha_0 + \omega} \\
& + \frac{1}{(\alpha_0 - \alpha_1)^2} (\alpha_0^2 (-\alpha_1 + \alpha_2 + \delta_1) + \alpha_0 (2\alpha_1^2 - 2\alpha_1\alpha_2 - \alpha_1\delta_0 - 2\alpha_1\delta_1 + \alpha_2\delta_0 + \delta_0\delta_1) - \alpha_1^3 + \alpha_1^2 (\alpha_2 + \delta_0 + \delta_1) - \alpha_1\alpha_2\delta_0 - \alpha_2\delta_0\delta_1) \cdot \frac{1}{\alpha_1 + \omega} \\
& + -\frac{\delta_1}{\alpha_0 - \alpha_1} (\alpha_1 - \alpha_2) (\alpha_0 - \alpha_1 + \delta_0) \cdot \frac{1}{(\alpha_1 + \omega)^2}
\end{aligned}$$

$$\begin{aligned}
L_{aaa}^{012} = & 1 \\
& + -\alpha_0 + \alpha_2 + \delta_0 + \delta_1 \cdot \frac{1}{\alpha_0 + \omega} \\
& + \delta_0\delta_1 (-\alpha_0 + \alpha_2) \cdot \frac{1}{(\alpha_0 + \omega)^3} \\
& + \alpha_0 (-\delta_0 - \delta_1) + \alpha_2 (\delta_0 + \delta_1) + \delta_0\delta_1 \cdot \frac{1}{(\alpha_0 + \omega)^2}
\end{aligned}$$

$$2.2 \quad L^{021} = \mathcal{F}^{-1}\left(\frac{s_0 s_2}{s_1}\right)$$

$$\begin{aligned} L_{abc}^{021} = & 1 \\ & + \frac{\delta_0 (\alpha_0 - \alpha_1) (-\alpha_0 + \alpha_2 + \delta_2)}{(\alpha_0 - \alpha_2) (-\alpha_0 + \alpha_1 + \delta_1)} \cdot \frac{1}{\alpha_0 + \omega} \\ & + \frac{\delta_2 (\alpha_1 - \alpha_2) (\alpha_0 - \alpha_2 + \delta_0)}{(\alpha_0 - \alpha_2) (\alpha_1 - \alpha_2 + \delta_1)} \cdot \frac{1}{\alpha_2 + \omega} \\ & + \frac{\delta_1 (\alpha_0 - \alpha_1 + \delta_0 - \delta_1) (\alpha_1 - \alpha_2 + \delta_1 - \delta_2)}{(-\alpha_0 + \alpha_1 + \delta_1) (\alpha_1 - \alpha_2 + \delta_1)} \cdot \frac{1}{\alpha_1 + \delta_1 + \omega} \end{aligned}$$

$$\begin{aligned} L_{aac}^{021} = & 1 \\ & + \frac{1}{(-\alpha_0 + \alpha_1 + \delta_1)^2} (\alpha_0^2 (\delta_0 + \delta_2) + \alpha_0 (-2\alpha_1 \delta_0 - 2\alpha_1 \delta_2 - \delta_0 \delta_1 - \delta_1 \delta_2) + \alpha_1^2 (\delta_0 + \delta_2) + \alpha_1 (\delta_0 \delta_1 + \delta_1 \delta_2) + \delta_0 \delta_1 \delta_2) \cdot \frac{1}{\alpha_0 + \omega} \\ & + \frac{\delta_0 \delta_2 (\alpha_0 - \alpha_1)}{\alpha_0 - \alpha_1 - \delta_1} \cdot \frac{1}{(\alpha_0 + \omega)^2} \\ & + -\frac{\delta_1}{(-\alpha_0 + \alpha_1 + \delta_1)^2} (\alpha_0 - \alpha_1 + \delta_0 - \delta_1) (\alpha_0 - \alpha_1 - \delta_1 + \delta_2) \cdot \frac{1}{\alpha_1 + \delta_1 + \omega} \end{aligned}$$

$$\begin{aligned} L_{aba}^{021} = & 1 \\ & + \frac{1}{(\alpha_0 - \alpha_2)^2} (-\alpha_0^3 + \alpha_0^2 (\alpha_1 + 2\alpha_2 + \delta_0 + \delta_2) + \alpha_0 (-2\alpha_1 \alpha_2 - \alpha_1 \delta_2 - \alpha_2^2 - 2\alpha_2 \delta_0 - \alpha_2 \delta_2) + \alpha_1 (\alpha_2^2 + \alpha_2 \delta_2 - \delta_0 \delta_2) + \alpha_2^2 \delta_0 + \alpha_2 \delta_0 \delta_2) \cdot \frac{1}{\alpha_0 + \omega} \\ & + \frac{\delta_0}{\alpha_0 - \alpha_2} (\alpha_0 - \alpha_1) (-\alpha_0 + \alpha_2 + \delta_2) \cdot \frac{1}{(\alpha_0 + \omega)^2} \\ & + \frac{\delta_2}{(\alpha_0 - \alpha_2)^2} (\alpha_1 - \alpha_2) (\alpha_0 - \alpha_2 + \delta_0) \cdot \frac{1}{\alpha_2 + \omega} \end{aligned}$$

$$\begin{aligned} L_{abb}^{021} = & 1 \\ & + -\frac{\delta_0}{(\alpha_0 - \alpha_2)^2} (\alpha_0 - \alpha_1) (-\alpha_0 + \alpha_2 + \delta_2) \cdot \frac{1}{\alpha_0 + \omega} \\ & + \frac{1}{(\alpha_0 - \alpha_2)^2} (\alpha_0^2 (\alpha_1 - \alpha_2 + \delta_2) + \alpha_0 (-2\alpha_1 \alpha_2 + \alpha_1 \delta_0 + 2\alpha_2^2 - \alpha_2 \delta_0 - 2\alpha_2 \delta_2 + \delta_0 \delta_2) + \alpha_1 (\alpha_2^2 - \alpha_2 \delta_0 - \delta_0 \delta_2) - \alpha_2^3 + \alpha_2^2 (\delta_0 + \delta_2)) \cdot \frac{1}{\alpha_2 + \omega} \\ & + \frac{\delta_2}{\alpha_0 - \alpha_2} (\alpha_1 - \alpha_2) (\alpha_0 - \alpha_2 + \delta_0) \cdot \frac{1}{(\alpha_2 + \omega)^2} \end{aligned}$$

$$\begin{aligned}
L_{aaa}^{021} = & 1 \\
& + (-\alpha_0 + \alpha_1 + \delta_0 + \delta_2) \cdot \frac{1}{\alpha_0 + \omega} \\
& + \delta_0 \delta_2 (-\alpha_0 + \alpha_1) \cdot \frac{1}{(\alpha_0 + \omega)^3} \\
& + \alpha_0 (-\delta_0 - \delta_2) + \alpha_1 (\delta_0 + \delta_2) + \delta_0 \delta_2 \cdot \frac{1}{(\alpha_0 + \omega)^2}
\end{aligned}$$

2.3 $L^{120} = \mathcal{F}^{-1}\left(\frac{s_1 s_2}{s_0}\right)$

$$\begin{aligned}
L_{abc}^{120} = & 1 \\
& + -\frac{\delta_1 (\alpha_0 - \alpha_1) (-\alpha_1 + \alpha_2 + \delta_2)}{(\alpha_1 - \alpha_2) (\alpha_0 - \alpha_1 + \delta_0)} \cdot \frac{1}{\alpha_1 + \omega} \\
& + \frac{\delta_2 (\alpha_0 - \alpha_2) (\alpha_1 - \alpha_2 + \delta_1)}{(\alpha_1 - \alpha_2) (\alpha_0 - \alpha_2 + \delta_0)} \cdot \frac{1}{\alpha_2 + \omega} \\
& + -\frac{\delta_0 (\alpha_0 - \alpha_1 + \delta_0 - \delta_1) (\alpha_0 - \alpha_2 + \delta_0 - \delta_2)}{(\alpha_0 - \alpha_1 + \delta_0) (\alpha_0 - \alpha_2 + \delta_0)} \cdot \frac{1}{\alpha_0 + \delta_0 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{aac}^{120} = & 1 \\
& + \frac{1}{(\alpha_0 - \alpha_1 + \delta_0)^2} (\alpha_0^2 (\delta_1 + \delta_2) + \alpha_0 (-2\alpha_1 \delta_1 - 2\alpha_1 \delta_2 + \delta_0 \delta_1 + \delta_0 \delta_2) + \alpha_1^2 (\delta_1 + \delta_2) + \alpha_1 (-\delta_0 \delta_1 - \delta_0 \delta_2) + \delta_0 \delta_1 \delta_2) \cdot \frac{1}{\alpha_1 + \omega} \\
& + \frac{\delta_1 \delta_2 (\alpha_0 - \alpha_1)}{\alpha_0 - \alpha_1 + \delta_0} \cdot \frac{1}{(\alpha_1 + \omega)^2} \\
& + -\frac{\delta_0}{(\alpha_0 - \alpha_1 + \delta_0)^2} (\alpha_0 - \alpha_1 + \delta_0 - \delta_1) (\alpha_0 - \alpha_1 + \delta_0 - \delta_2) \cdot \frac{1}{\alpha_0 + \delta_0 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{aba}^{120} = & 1 \\
& + \frac{1}{(\alpha_1 - \alpha_2)^2} (\alpha_0 (\alpha_1^2 - 2\alpha_1 \alpha_2 - \alpha_1 \delta_2 + \alpha_2^2 + \alpha_2 \delta_2 - \delta_1 \delta_2) - \alpha_1^3 + \alpha_1^2 (2\alpha_2 + \delta_1 + \delta_2) + \alpha_1 (-\alpha_2^2 - 2\alpha_2 \delta_1 - \alpha_2 \delta_2) + \alpha_2^2 \delta_1 + \alpha_2 \delta_1 \delta_2) \cdot \frac{1}{\alpha_1 + \omega} \\
& + -\frac{\delta_1}{\alpha_1 - \alpha_2} (\alpha_0 - \alpha_1) (-\alpha_1 + \alpha_2 + \delta_2) \cdot \frac{1}{(\alpha_1 + \omega)^2} \\
& + \frac{\delta_2}{(\alpha_1 - \alpha_2)^2} (\alpha_0 - \alpha_2) (\alpha_1 - \alpha_2 + \delta_1) \cdot \frac{1}{\alpha_2 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{abb}^{120} = & 1 \\
& + \frac{\delta_1}{(\alpha_1 - \alpha_2)^2} (\alpha_0 - \alpha_1) (-\alpha_1 + \alpha_2 + \delta_2) \cdot \frac{1}{\alpha_1 + \omega} \\
& + \frac{1}{(\alpha_1 - \alpha_2)^2} (\alpha_0 (\alpha_1^2 - 2\alpha_1\alpha_2 + \alpha_1\delta_1 + \alpha_2^2 - \alpha_2\delta_1 - \delta_1\delta_2) + \alpha_1^2 (-\alpha_2 + \delta_2) + \alpha_1 (2\alpha_2^2 - \alpha_2\delta_1 - 2\alpha_2\delta_2 + \delta_1\delta_2) - \alpha_2^3 + \alpha_2^2 (\delta_1 + \delta_2)) \cdot \frac{1}{\alpha_2 + \omega} \\
& + \frac{\delta_2}{\alpha_1 - \alpha_2} (\alpha_0 - \alpha_2) (\alpha_1 - \alpha_2 + \delta_1) \cdot \frac{1}{(\alpha_2 + \omega)^2}
\end{aligned}$$

$$\begin{aligned}
L_{aaa}^{120} = & 1 \\
& + \alpha_0 - \alpha_1 + \delta_1 + \delta_2 \cdot \frac{1}{\alpha_1 + \omega} \\
& + \delta_1\delta_2 (\alpha_0 - \alpha_1) \cdot \frac{1}{(\alpha_1 + \omega)^3} \\
& + \alpha_0 (\delta_1 + \delta_2) + \alpha_1 (-\delta_1 - \delta_2) + \delta_1\delta_2 \cdot \frac{1}{(\alpha_1 + \omega)^2}
\end{aligned}$$

2.4 $L^{01x} = \mathcal{F}^{-1}(s_0 s_1)$

$$\begin{aligned}
L_{ab}^{01x} = & 1 \\
& + \frac{\delta_0 (\alpha_0 - \alpha_1 - \delta_1)}{\alpha_0 - \alpha_1} \cdot \frac{1}{\alpha_0 + \omega} \\
& + \frac{\delta_1 (\alpha_0 - \alpha_1 + \delta_0)}{\alpha_0 - \alpha_1} \cdot \frac{1}{\alpha_1 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{aa}^{01x} = & 1 \\
& + \delta_0 + \delta_1 \cdot \frac{1}{\alpha_0 + \omega} \\
& + \delta_0\delta_1 \cdot \frac{1}{(\alpha_0 + \omega)^2}
\end{aligned}$$

2.5 $L^{02x} = \mathcal{F}^{-1}(s_0 s_2)$

$$\begin{aligned}
L_{ab}^{02x} = & 1 \\
& + \frac{\delta_0 (\alpha_0 - \alpha_2 - \delta_2)}{\alpha_0 - \alpha_2} \cdot \frac{1}{\alpha_0 + \omega} \\
& + \frac{\delta_2 (\alpha_0 - \alpha_2 + \delta_0)}{\alpha_0 - \alpha_2} \cdot \frac{1}{\alpha_2 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{aa}^{02x} = & 1 \\
& + \delta_0 + \delta_2 \cdot \frac{1}{\alpha_0 + \omega} \\
& + \delta_0 \delta_2 \cdot \frac{1}{(\alpha_0 + \omega)^2}
\end{aligned}$$

$$\mathbf{2.6} \quad L^{12x} = \mathcal{F}^{-1}(s_1 s_2)$$

$$\begin{aligned}
L_{ab}^{12x} = & 1 \\
& + \frac{\delta_1 (\alpha_1 - \alpha_2 - \delta_2)}{\alpha_1 - \alpha_2} \cdot \frac{1}{\alpha_1 + \omega} \\
& + \frac{\delta_2 (\alpha_1 - \alpha_2 + \delta_1)}{\alpha_1 - \alpha_2} \cdot \frac{1}{\alpha_2 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{aa}^{12x} = & 1 \\
& + \delta_1 + \delta_2 \cdot \frac{1}{\alpha_1 + \omega} \\
& + \delta_1 \delta_2 \cdot \frac{1}{(\alpha_1 + \omega)^2}
\end{aligned}$$

$$\mathbf{2.7} \quad L^{0x2} = \mathcal{F}^{-1}\left(\frac{s_0}{s_x}\right)$$

$$\begin{aligned}
L_{ab}^{0x2} = & 1 \\
& + \frac{\delta_0 (\alpha_0 - \alpha_2)}{\alpha_0 - \alpha_2 - \delta_2} \cdot \frac{1}{\alpha_0 + \omega} \\
& + \frac{\delta_2 (\alpha_0 - \alpha_2 + \delta_0 - \delta_2)}{-\alpha_0 + \alpha_2 + \delta_2} \cdot \frac{1}{\alpha_2 + \delta_2 + \omega}
\end{aligned}$$

$$\begin{aligned}
L_{aa}^{0x2} = & 1 \\
& + -\alpha_0 + \alpha_2 + \delta_0 \cdot \frac{1}{\alpha_0 + \omega} \\
& + \delta_0 (-\alpha_0 + \alpha_2) \cdot \frac{1}{(\alpha_0 + \omega)^2}
\end{aligned}$$

$$2.8 \quad L^{0x1} = \mathcal{F}^{-1}\left(\frac{s_0}{s_x}\right)$$

$$\begin{aligned} L_{ab}^{0x1} = & 1 \\ & + \frac{\delta_0 (\alpha_0 - \alpha_1)}{\alpha_0 - \alpha_1 - \delta_1} \cdot \frac{1}{\alpha_0 + \omega} \\ & + \frac{\delta_1 (\alpha_0 - \alpha_1 + \delta_0 - \delta_1)}{-\alpha_0 + \alpha_1 + \delta_1} \cdot \frac{1}{\alpha_1 + \delta_1 + \omega} \end{aligned}$$

$$\begin{aligned} L_{aa}^{0x1} = & 1 \\ & + -\alpha_0 + \alpha_1 + \delta_0 \cdot \frac{1}{\alpha_0 + \omega} \\ & + \delta_0 (-\alpha_0 + \alpha_1) \cdot \frac{1}{(\alpha_0 + \omega)^2} \end{aligned}$$

$$2.9 \quad L^{1x0} = \mathcal{F}^{-1}\left(\frac{s_1}{s_x}\right)$$

$$\begin{aligned} L_{ab}^{1x0} = & 1 \\ & + \frac{\delta_1 (\alpha_0 - \alpha_1)}{\alpha_0 - \alpha_1 + \delta_0} \cdot \frac{1}{\alpha_1 + \omega} \\ & + -\frac{\delta_0 (\alpha_0 - \alpha_1 + \delta_0 - \delta_1)}{\alpha_0 - \alpha_1 + \delta_0} \cdot \frac{1}{\alpha_0 + \delta_0 + \omega} \end{aligned}$$

$$\begin{aligned} L_{aa}^{1x0} = & 1 \\ & + \alpha_0 - \alpha_1 + \delta_1 \cdot \frac{1}{\alpha_1 + \omega} \\ & + \delta_1 (\alpha_0 - \alpha_1) \cdot \frac{1}{(\alpha_1 + \omega)^2} \end{aligned}$$

3 Calcul des termes en L pour le couplage

$$3.1 \quad \text{Terme : } \mathcal{F}^{-1}(s_0)$$

$$\begin{aligned} L^0 = & 1 \\ & + \delta_0 \cdot \frac{1}{\alpha_0 + \omega} \end{aligned}$$

3.2 Terme ab : $\mathcal{F}^{-1}(s_0 s_1)$

$$\begin{aligned} L_{ab}^{01} = & 1 \\ & + \frac{\delta_0 (\alpha_0 - \alpha_1 - \delta_1)}{\alpha_0 - \alpha_1} \cdot \frac{1}{\alpha_0 + \omega} \\ & + \frac{\delta_1 (\alpha_0 - \alpha_1 + \delta_0)}{\alpha_0 - \alpha_1} \cdot \frac{1}{\alpha_1 + \omega} \end{aligned}$$

3.3 Terme aa : $\mathcal{F}^{-1}(s_0 s_1)$

$$\begin{aligned} L_{aa}^{01} = & 1 \\ & + \delta_0 + \delta_1 \cdot \frac{1}{\alpha_0 + \omega} \\ & + \delta_0 \delta_1 \cdot \frac{1}{(\alpha_0 + \omega)^2} \end{aligned}$$

3.4 Terme : $\mathcal{F}^{-1}(\omega^2 s_0)$

$$\begin{aligned} L^0 = & -\alpha_0 \delta_0 + \delta_0 \omega + \omega^2 \\ & + \alpha_0^2 \delta_0 \cdot \frac{1}{\alpha_0 + \omega} \end{aligned}$$

3.5 Terme ab : $\mathcal{F}^{-1}(\omega^2 s_0 s_1)$

$$\begin{aligned} L_{ab}^{01} = & -\alpha_0 \delta_0 - \alpha_1 \delta_1 + \delta_0 \delta_1 + \omega^2 + \omega (\delta_0 + \delta_1) \\ & + \frac{\alpha_0^2 \delta_0}{\alpha_0 - \alpha_1} (\alpha_0 - \alpha_1 - \delta_1) \cdot \frac{1}{\alpha_0 + \omega} \\ & + \frac{\alpha_1^2 \delta_1}{\alpha_0 - \alpha_1} (\alpha_0 - \alpha_1 + \delta_0) \cdot \frac{1}{\alpha_1 + \omega} \end{aligned}$$

3.6 Terme aa : $\mathcal{F}^{-1}(\omega^2 s_0 s_1)$

$$\begin{aligned} L_{aa}^{01} = & -\alpha_0 \delta_0 - \alpha_0 \delta_1 + \delta_0 \delta_1 + \omega^2 + \omega (\delta_0 + \delta_1) \\ & + \alpha_0 (\alpha_0 (\delta_0 + \delta_1) - 2\delta_0 \delta_1) \cdot \frac{1}{\alpha_0 + \omega} \\ & + \alpha_0^2 \delta_0 \delta_1 \cdot \frac{1}{(\alpha_0 + \omega)^2} \end{aligned}$$